

Lab: Data Manipulation and Cleaning

Actuarial Data Science - Open Learning Resource

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Learning Objectives

- Learn how to import data, check data quality, and clean data.
- Learn how to manipulate and transform data.

Case Study A - French Insurance Dataset

We will continue to use the `freMTPL2freq` dataset. As a preview, this dataset includes risk features collected for 677,991 motor third-party liability policies, observed mostly over one year. In addition, `freMTPL2freq` contains both the risk features and the claim number per policy. The dataset consists of 12 columns:

- `IDpol`: The policy ID (used to link with the claims dataset).
- `ClaimNb`: Number of claims during the exposure period.
- `Exposure`: The period of exposure for a policy, in years.
- `Area`: The area code.
- `VehPower`: The power of the car (ordered categorical).
- `VehAge`: The vehicle age, in years.
- `DrivAge`: The driver age, in years (in France, people can drive a car at 18).
- `BonusMalus`: Bonus/malus, between 50 and 350 (<100 means bonus, >100 means malus in France).
- `VehBrand`: The car brand (unknown categories).
- `VehGas`: The fuel type (Diesel or regular).
- `Density`: The density of inhabitants (number of inhabitants per km²) in the city where the driver lives.
- `Region`: The policy regions in France (based on a standard French classification).

Let's first import the data, and then begin by briefly examining it.

```
library(CASdatasets)
library(tidyverse)
```

```
data(freMTPL2freq)
```

```
str(freMTPL2freq)
```

```
'data.frame':  678013 obs. of  12 variables:
 $ IDpol      : num  1 3 5 10 11 13 15 17 18 21 ...
 $ ClaimNb    : 'table' num [1:678013(1d)] 1 1 1 1 1 1 1 1 1 ...
 $ Exposure   : num  0.1 0.77 0.75 0.09 0.84 0.52 0.45 0.27 0.71 0.15 ...
 $ VehPower   : int   5 5 6 7 7 6 6 7 7 7 ...
 $ VehAge     : int   0 0 2 0 0 2 2 0 0 0 ...
 $ DrivAge    : int  55 55 52 46 46 38 38 33 33 41 ...
```

```

$ BonusMalus: int  50 50 50 50 50 50 50 68 68 50 ...
$ VehBrand   : Factor w/ 11 levels "B1","B10","B11",...: 4 4 4 4 4 4 4 4 4 ...
$ VehGas      : chr  "Regular" "Regular" "Diesel" "Diesel" ...
$ Area        : Factor w/ 6 levels "A","B","C","D",...: 4 4 2 2 2 5 5 3 3 2 ...
$ Density     : int  1217 1217 54 76 76 3003 3003 137 137 60 ...
$ Region      : Factor w/ 21 levels "Alsace","Aquitaine",...: 21 21 18 2 2 16 16 13 13 17 ...

```

```
summary(freMTPL2freq)
```

IDpol	ClaimNb	Exposure	VehPower
Min. : 1	n.vars :1	Min. :0.002732	Min. : 4.000
1st Qu.:1157951	n.cases:36102	1st Qu.:0.180000	1st Qu.: 5.000
Median :2272152		Median :0.490000	Median : 6.000
Mean :2621857		Mean :0.528750	Mean : 6.455
3rd Qu.:4046274		3rd Qu.:0.990000	3rd Qu.: 7.000
Max. :6114330		Max. :2.010000	Max. :15.000

VehAge	DrivAge	BonusMalus	VehBrand
Min. : 0.000	Min. : 18.0	Min. : 50.00	B12 :166024
1st Qu.: 2.000	1st Qu.: 34.0	1st Qu.: 50.00	B1 :162736
Median : 6.000	Median : 44.0	Median : 50.00	B2 :159861
Mean : 7.044	Mean : 45.5	Mean : 59.76	B3 : 53395
3rd Qu.: 11.000	3rd Qu.: 55.0	3rd Qu.: 64.00	B5 : 34753
Max. :100.000	Max. :100.0	Max. :230.00	B6 : 28548
			(Other): 72696

VehGas	Area	Density
Length:678013	A:103957	Min. : 1
Class :character	B: 75459	1st Qu.: 92
Mode :character	C:191880	Median : 393
	D:151596	Mean : 1792
	E:137167	3rd Qu.: 1658
	F: 17954	Max. :27000

Region
Centre :160601
Rhone-Alpes : 84752
Provence-Alpes-Cotes-D'Azur: 79315
Ile-de-France : 69791
Bretagne : 42122
Nord-Pas-de-Calais : 40275
(Other) :201157

From the outputs above, we can see that there are 678013 individual car insurance policies and 12 variables associated with each policy. At first glance, we notice that the data types of some columns may need adjustment. For example, ClaimNb is stored as a table and VehGas is stored as a character variable. We may want to convert these to integer and factor, respectively. However, note that some modelling packages can handle these automatically.

```

# Convert ClaimNb from a table to integer
freMTPL2freq$ClaimNb <- as.integer(as.numeric(freMTPL2freq$ClaimNb))

# Convert VehGas from character to factor
freMTPL2freq$VehGas <- as.factor(freMTPL2freq$VehGas)

# Recheck the data structure after the adjustments

```

```
# str(freMTPL2freq)
# summary(freMTPL2freq)
```

Tasks for Case Study A

Based on the dataset above, complete the following tasks:

1. Are there any missing values in the dataset? How can you check this?
2. Examine the distribution of exposure and the number of claims. Do you observe any unusual patterns?
3. Is Area an ordinal categorical variable? How can you verify this?
4. Explore the relationship between driver age (DrivAge) and claim frequency. How does age influence the frequency of claims?
5. Analyse the relationships between the predictors. Are there any strong correlations or dependencies? What are the potential implications for modelling?

Solution A.1: Missing values

Task:

Are there any missing values in the dataset? How can you check this?

```
# Check for NA values in freMTPL2freq
na_summary_freq <- sapply(freMTPL2freq, function(x) sum(is.na(x)))
print(na_summary_freq)
```

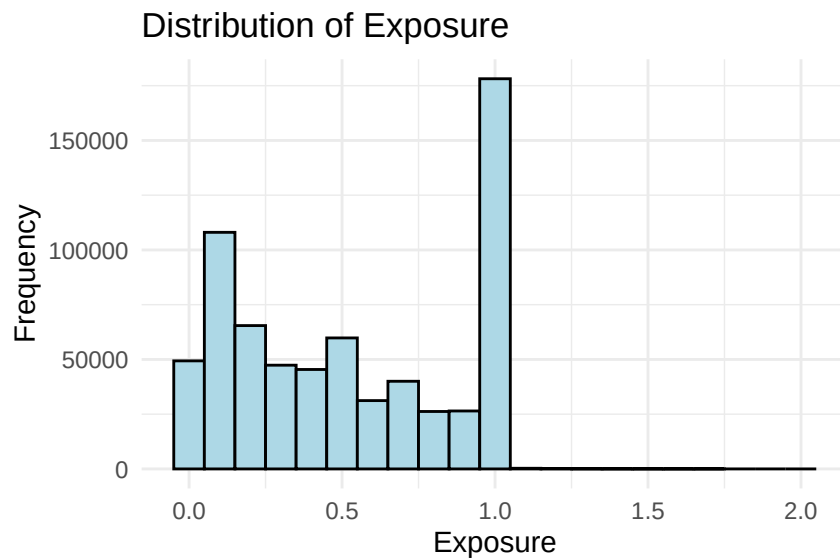
IDpol	ClaimNb	Exposure	VehPower	VehAge	DrivAge	BonusMalus
0	0	0	0	0	0	0
VehBrand	VehGas	Area	Density	Region		
0	0	0	0	0		

Fortunately, there are no missing values in this dataset.

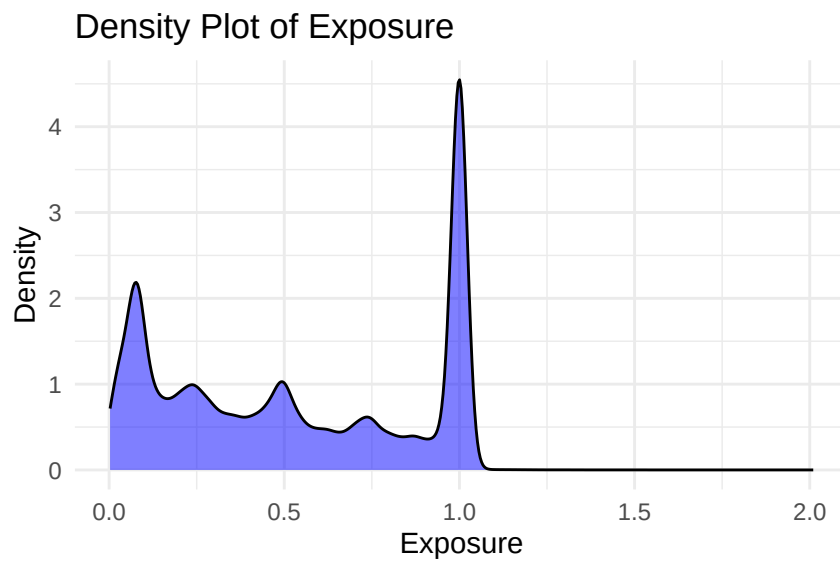
Solution A.2: Distribution of exposure and claims

Task: Examine the distribution of exposure and the number of claims. Do you observe any unusual patterns?

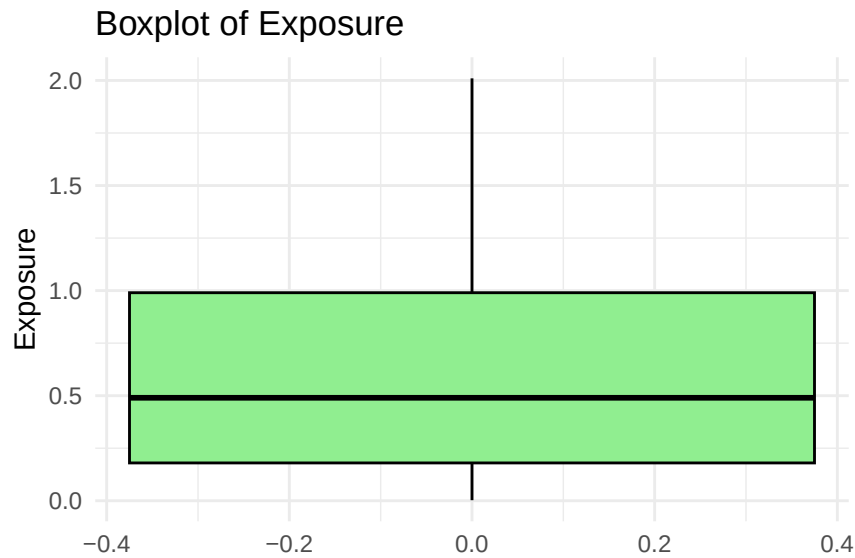
```
# Histogram of exposure
ggplot(freMTPL2freq, aes(x = Exposure)) +
  geom_histogram(binwidth = 0.1, fill = "lightblue", colour = "black") +
  labs(title = "Distribution of Exposure", x = "Exposure", y = "Frequency") +
  theme_minimal()
```



```
# Density plot of exposure
ggplot(freMTPL2freq, aes(x = Exposure)) +
  geom_density(fill = "blue", alpha = 0.5) +
  labs(title = "Density Plot of Exposure", x = "Exposure", y = "Density") +
  theme_minimal()
```



```
# Boxplot of exposure
ggplot(freMTPL2freq, aes(y = Exposure)) +
  geom_boxplot(fill = "lightgreen", colour = "black") +
  labs(title = "Boxplot of Exposure", y = "Exposure") +
  theme_minimal()
```



```
# Frequency table of the number of claims
freMTPL2freq %>%
  count(ClaimNb) %>%
  print()
```

	ClaimNb	n
1	0	643953
2	1	32178
3	2	1784
4	3	82
5	4	7
6	5	2
7	6	1
8	8	1
9	9	1
10	11	3
11	16	1

We consider several plots to examine the distribution of exposure. Typically, you would only need to show one of these in an EDA. Note that some exposures are greater than one year (i.e., 1224 policies).

In addition, we present the frequency table of the number of claims. There are only 9 policies with more than 4 claims, as shown in the table. Without further information, it is difficult to determine whether these entries are errors. You may choose to keep them or consider capping them. For example, in Noll, Salzmann, and Wuthrich (2020), all exposures greater than 1 are set to 1, and all claim numbers greater than 4 are set to 4.

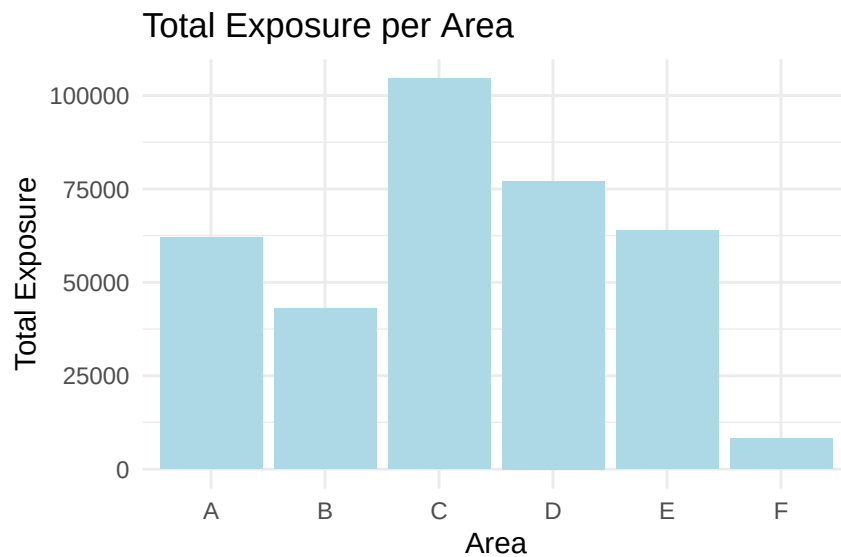
Solution A.3: Is Area ordinal?

Task: Is Area an ordinal categorical variable? How can you verify this?

```
# Calculate total exposure per area code
total_exposure_per_area <- freMTPL2freq %>%
  group_by(Area) %>%
  summarise(TotalExposure = sum(Exposure, na.rm = TRUE))

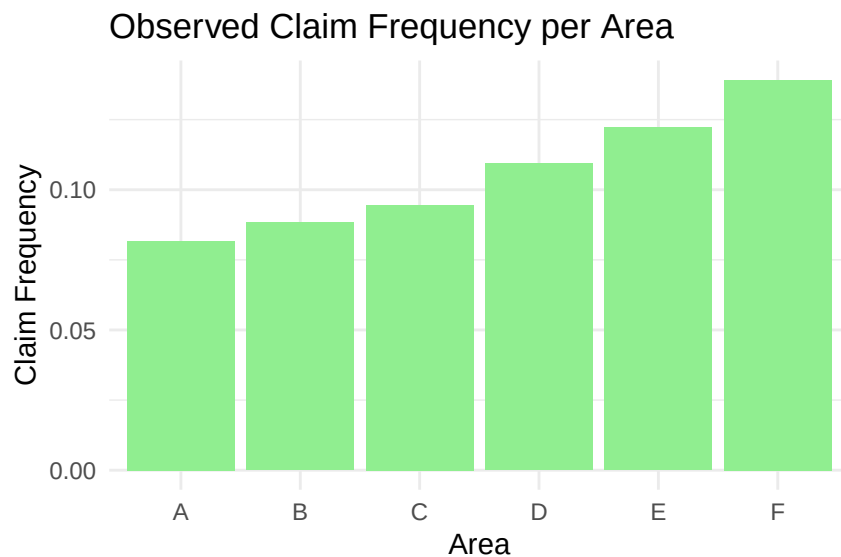
# Bar plot of total exposure per area code
```

```
ggplot(total_exposure_per_area, aes(x = Area, y = TotalExposure)) +
  geom_col(fill = "lightblue") +
  labs(title = "Total Exposure per Area", x = "Area", y = "Total Exposure") +
  theme_minimal()
```



```
# Calculate claim frequency per area code
claim_frequency_per_area <- freMTPL2freq %>%
  group_by(Area) %>%
  summarise(
    TotalClaims = sum(ClaimNb, na.rm = TRUE),
    TotalExposure = sum(Exposure, na.rm = TRUE),
    ClaimFrequency = TotalClaims / TotalExposure
  )
```

```
# Bar plot of claim frequency per area code
ggplot(claim_frequency_per_area, aes(x = Area, y = ClaimFrequency)) +
  geom_col(fill = "lightgreen") +
  labs(title = "Observed Claim Frequency per Area", x = "Area", y = "Claim Frequency") +
  theme_minimal()
```



We first examine whether the total exposure is roughly the same across areas. This is not the case; for example, Area F has a much lower total exposure.

Next, we examine the observed claim frequency by area. The claim frequency increases consistently from Area A to Area F, suggesting that Area has a natural ordering. Therefore, Area can be treated as an ordinal categorical variable.

💡 Exercise:

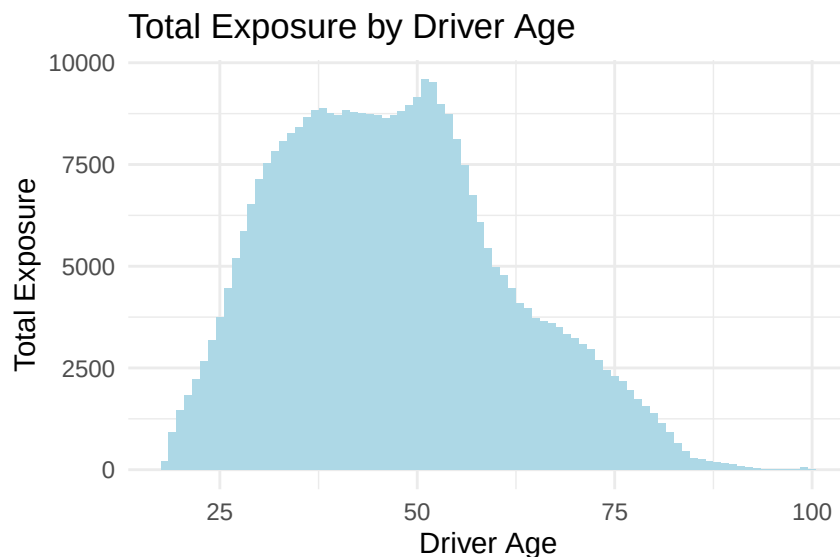
Is VehPower an ordinal variable? Can you follow the code above to check this?

Solution A.4: Age vs claim frequency

Task: Explore the relationship between driver age (DrvAge) and claim frequency. How does age influence the frequency of claims?

```
# Calculate total exposure by driver age
total_exposure_per_age <- freMTPL2freq %>%
  group_by(DrvAge) %>%
  summarise(TotalExposure = sum(Exposure, na.rm = TRUE)) %>%
  arrange(DrvAge)

# Bar plot of total exposure by driver age
ggplot(total_exposure_per_age, aes(x = DrvAge, y = TotalExposure)) +
  geom_col(fill = "lightblue") +
  labs(title = "Total Exposure by Driver Age", x = "Driver Age", y = "Total Exposure") +
  theme_minimal()
```



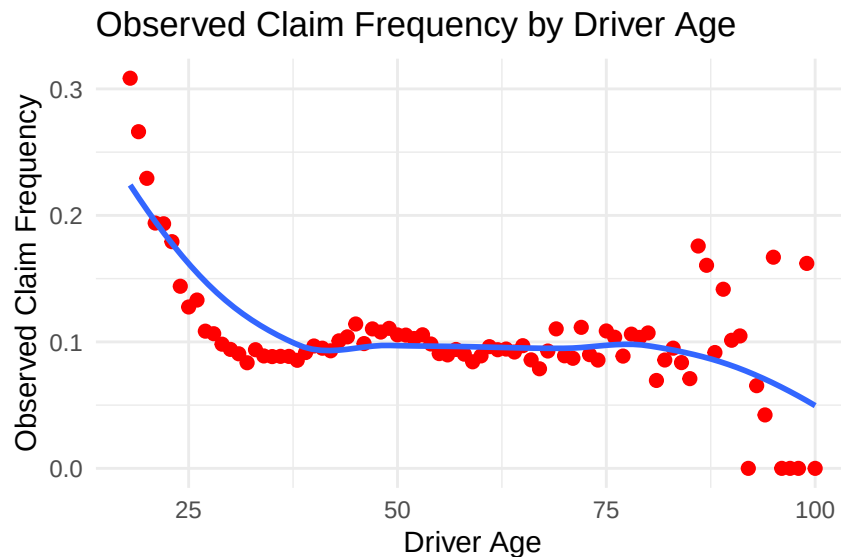
```
# Calculate observed claim frequency by driver age
observed_frequency_per_age <- freMTPL2freq %>%
  group_by(DrvAge) %>%
  summarise(
    TotalClaims = sum(ClaimNb, na.rm = TRUE),
    TotalExposure = sum(Exposure, na.rm = TRUE),
    ObservedFrequency = TotalClaims / TotalExposure
  ) %>%
```

```

arrange(DrivAge)

# Plot observed claim frequency by driver age
ggplot(observed_frequency_per_age, aes(x = DrivAge, y = ObservedFrequency)) +
  geom_point(colour = "red", size = 2) +
  geom_smooth(se = FALSE) +
  labs(title = "Observed Claim Frequency by Driver Age", x = "Driver Age", y = "Observed Claim Frequency") +
  theme_minimal()

```



From the plots above, we observe that the relationship between the predictor `DrivAge` and the observed claim frequency is non-linear. Please note this, as we will explore how to incorporate this into modelling in the coming weeks.

💡 Exercise:

Can you follow the code above, or write your own code, to explore the relationship between the observed claim frequency and other predictors in the dataset? Did you find any interesting patterns?

Solution A.5: Relationships between predictors

Task: Analyse the relationships between the predictors. Are there any strong correlations or dependencies? What are the potential implications for modelling?

```

# Convert Area to an ordered numeric variable (for exploratory purposes only)
freMTPL2freq$AreaNumeric <- as.numeric(as.ordered(freMTPL2freq$Area))

# Select relevant variables
correlation_data <- freMTPL2freq %>%
  select(AreaNumeric, VehPower, VehAge, DrivAge, BonusMalus, Density)

# Compute the Pearson correlation matrix
correlation_matrix <- cor(correlation_data, method = "pearson")

# Display the correlation matrix
print(correlation_matrix)

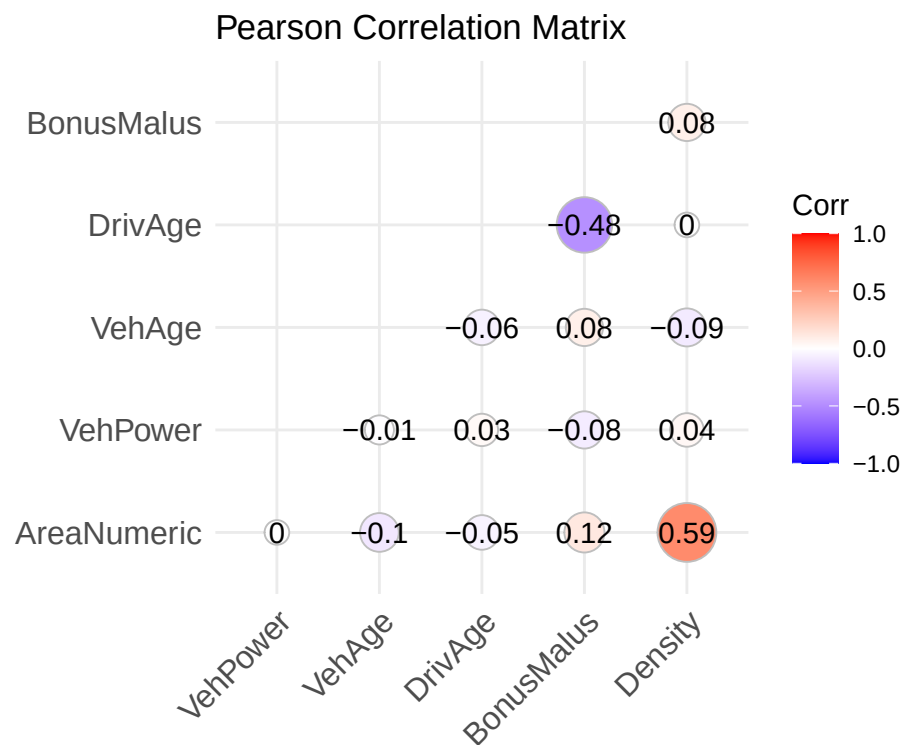
```


	AreaNumeric	VehPower	VehAge	DrivAge	BonusMalus
AreaNumeric	1.000000000	0.003176694	-0.104530220	-0.045180127	0.12085798
VehPower	0.003176694	1.000000000	-0.006001487	0.030107579	-0.07589469
VehAge	-0.104530220	-0.006001487	1.000000000	-0.059213383	0.07992307
DrivAge	-0.045180127	0.030107579	-0.059213383	1.000000000	-0.47996604
BonusMalus	0.120857981	-0.075894688	0.079923071	-0.479966037	1.000000000
Density	0.589375413	0.042900681	-0.090427830	-0.004699793	0.07771679

	Density
AreaNumeric	0.589375413
VehPower	0.042900681
VehAge	-0.090427830
DrivAge	-0.004699793
BonusMalus	0.077716791
Density	1.000000000

```
# Load package for visualisation
library(ggcorrplot)

# Visualise the correlation matrix
ggcorrplot(
  correlation_matrix,
  method = "circle",
  type = "lower",
  lab = TRUE,
  title = "Pearson Correlation Matrix"
)
```



Here, we examine the correlations between numerical and ordinal features. Note that Area has been converted to an ordered numeric variable for exploratory purposes, so the resulting correlations should be interpreted with caution.

From the correlation matrix, we observe that Area and Density appear to have a relatively strong positive association. In addition, there is a negative relationship between DriveAge and BonusMalus.

Examining relationships between predictors is important because it helps identify potential multicollinearity, reveals possible interactions, and provides insights into how predictors may jointly influence the response variable.

Exercise:

In the above, we only considered Pearson's correlation between numerical features. Can you explore additional interrelationships between predictors? For example, you might investigate how vehicle brand relates to other vehicle characteristics, or to driver and policy characteristics.

For reference, see [Noll, Salzmann, and Wuthrich \(2020\)](#) for some in-depth bivariate analysis in EDA for this dataset.

Case study B - Default of Credit Card Clients

In 2005, Taiwan experienced a credit card debt crisis, driven in part by aggressive expansion in the consumer credit market. Financial institutions issued a large number of credit cards, often to customers with limited repayment capacity. At the same time, many cardholders accumulated substantial debt through frequent credit use. This combination led to a deterioration in credit quality and weakened confidence in the consumer finance sector, posing significant challenges for both financial institutions and borrowers ([Yeh and Lien 2009](#)).

This episode highlights the importance of risk prediction in consumer finance. By analysing financial information—such as transaction records and repayment history—institutions can better assess credit risk and mitigate potential losses.

This dataset contains information on customers' default payments, with 30,000 observations described over 24 attributes. It includes default payment status, demographic factors, credit information, repayment history, and bill statements of credit card clients in Taiwan from April 2005 to September 2005. The data can be downloaded from the [UCI Machine Learning Repository](#).

This case study examines customers' default payments in Taiwan and compares the predictive accuracy of the probability of default using shrinkage techniques (lasso, ridge, and elastic net regression) and non-shrinkage methods such as logistic regression. The response variable is a binary variable indicating default payment (Yes = 1, No = 0). The dataset contains 23 explanatory variables:

- LIMIT_BAL: Amount of the given credit (NT dollar), including both individual and supplementary family credit.
- SEX: Gender (1 = male; 2 = female).
- EDUCATION: Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).
- MARRIAGE: Marital status (1 = married; 2 = single; 3 = others).
- AGE: Age (years).
- PAY_1 - PAY_6: History of past payments. We track the monthly repayment status from April to September 2005:
 - PAY_1: repayment status in September 2005
 - PAY_2: repayment status in August 2005
 - ...
 - PAY_6: repayment status in April 2005

The repayment status is coded as:

- -2: no consumption
 - -1: paid in full
 - 0: use of revolving credit
 - 1: payment delay for one month
 - ...
 - 9: payment delay for nine months or more
- BILL_AMT1 - BILL_AMT6: Amount of bill statement (NT dollar):
 - BILL_AMT1: bill amount in September 2005
 - BILL_AMT2: bill amount in August 2005
 - ...
 - BILL_AMT6: bill amount in April 2005
 - PAY_AMT1 - PAY_AMT6: Amount of previous payments (NT dollar):
 - PAY_AMT1: payment amount in September 2005
 - PAY_AMT2: payment amount in August 2005
 - ...
 - PAY_AMT6: payment amount in April 2005

Tasks for Case Study B

Based on the dataset above, complete the following tasks:

1. Are there any missing values in the data? If so, suggest possible methods for imputation and apply one of them.
2. Using visualisations, explore the predictor variables to understand their distributions as well as the relationships between predictors.
3. Are there any transformations of the predictors that might improve model performance?
4. Rename the column “default payment next month” as “default”. Are there strong relationships between the default variable and other numeric variables? How can you handle highly correlated variables?

Import data

This case study focuses on pre-processing the data before applying machine learning techniques to predict the response variable.

Load required packages

```
library(data.table)
library(readxl)
library(ggplot2)
library(tidyverse)
library(naniar)
library(corrplot)
library(caret)
library(gridExtra)
library(ggcorrplot)
library(glmnet)
```

Import data

```
data <- read_excel("credit.xls", skip = 1)
```

Examine the data structure

```
dim(data) # dimension of the data
```

```
[1] 30000    25
```

```
str(data) # structure of the data
```

```
tibble [30,000 x 25] (S3: tbl_df/tbl/data.frame)
```

```
$ ID           : num [1:30000] 1 2 3 4 5 6 7 8 9 10 ...
$ LIMIT_BAL    : num [1:30000] 20000 120000 90000 50000 50000 50000 500000 100000 140000 200000 ...
$ SEX          : num [1:30000] 2 2 2 2 1 1 1 2 2 1 ...
$ EDUCATION    : num [1:30000] 2 2 2 2 2 1 1 2 3 3 ...
$ MARRIAGE     : num [1:30000] 1 2 2 1 1 2 2 2 1 2 ...
$ AGE          : num [1:30000] 24 26 34 37 57 37 29 23 28 35 ...
$ PAY_0        : num [1:30000] 2 -1 0 0 -1 0 0 0 0 -2 ...
$ PAY_2        : num [1:30000] 2 2 0 0 0 0 0 -1 0 -2 ...
$ PAY_3        : num [1:30000] -1 0 0 0 -1 0 0 -1 2 -2 ...
$ PAY_4        : num [1:30000] -1 0 0 0 0 0 0 0 0 -2 ...
$ PAY_5        : num [1:30000] -2 0 0 0 0 0 0 0 0 -1 ...
$ PAY_6        : num [1:30000] -2 2 0 0 0 0 0 -1 0 -1 ...
$ BILL_AMT1    : num [1:30000] 3913 2682 29239 46990 8617 ...
$ BILL_AMT2    : num [1:30000] 3102 1725 14027 48233 5670 ...
$ BILL_AMT3    : num [1:30000] 689 2682 13559 49291 35835 ...
$ BILL_AMT4    : num [1:30000] 0 3272 14331 28314 20940 ...
$ BILL_AMT5    : num [1:30000] 0 3455 14948 28959 19146 ...
$ BILL_AMT6    : num [1:30000] 0 3261 15549 29547 19131 ...
$ PAY_AMT1     : num [1:30000] 0 0 1518 2000 2000 ...
$ PAY_AMT2     : num [1:30000] 689 1000 1500 2019 36681 ...
$ PAY_AMT3     : num [1:30000] 0 1000 1000 1200 10000 657 38000 0 432 0 ...
$ PAY_AMT4     : num [1:30000] 0 1000 1000 1100 9000 ...
$ PAY_AMT5     : num [1:30000] 0 0 1000 1069 689 ...
$ PAY_AMT6     : num [1:30000] 0 2000 5000 1000 679 ...
$ default payment next month: num [1:30000] 1 1 0 0 0 0 0 0 0 0 ...
```

Rename variables

```
colnames(data)[colnames(data) == "PAY_0"] <- "PAY_1"
colnames(data)[colnames(data) == "default payment next month"] <- "default"
```

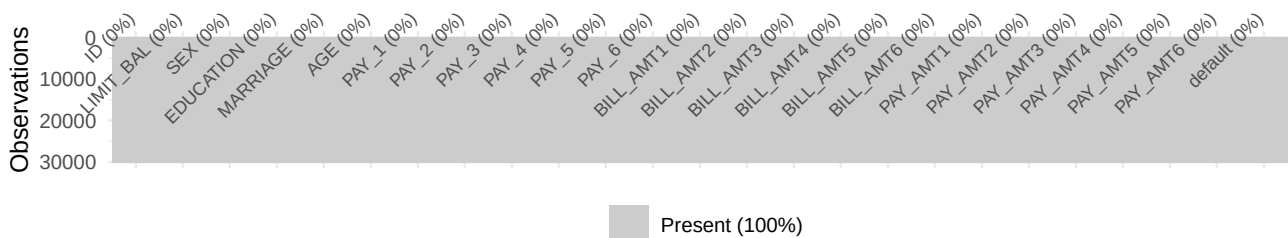
```
data$default <- as.factor(data$default)
data$SEX <- as.factor(data$SEX)
```

Solution B.1: Missing values

Task: Are there any missing values in the data? If so, suggest possible methods for imputation and apply one of them.

Checking missing values

```
vis_miss(data) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8))
```



```
colSums(is.na(data))
```

ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_1	PAY_2
0	0	0	0	0	0	0	0
PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4
0	0	0	0	0	0	0	0
BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6
0	0	0	0	0	0	0	0
default							
0							

```
summary(data)
```

ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE
Min. : 1	Min. : 10000	1:11888	Min. : 0.000	Min. : 0.000
1st Qu.: 7501	1st Qu.: 50000	2:18112	1st Qu.: 1.000	1st Qu.: 1.000
Median : 15000	Median : 140000		Median : 2.000	Median : 2.000
Mean : 15000	Mean : 167484		Mean : 1.853	Mean : 1.552
3rd Qu.: 22500	3rd Qu.: 240000		3rd Qu.: 2.000	3rd Qu.: 2.000
Max. : 30000	Max. : 1000000		Max. : 6.000	Max. : 3.000
AGE	PAY_1	PAY_2	PAY_3	
Min. : 21.00	Min. : -2.0000	Min. : -2.0000	Min. : -2.0000	
1st Qu.: 28.00	1st Qu.: -1.0000	1st Qu.: -1.0000	1st Qu.: -1.0000	
Median : 34.00	Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 35.49	Mean : -0.0167	Mean : -0.1338	Mean : -0.1662	

3rd Qu.:41.00	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :79.00	Max. : 8.0000	Max. : 8.0000	Max. : 8.0000
PAY_4	PAY_5	PAY_6	BILL_AMT1
Min. :-2.0000	Min. :-2.0000	Min. :-2.0000	Min. :-165580
1st Qu.: -1.0000	1st Qu.: -1.0000	1st Qu.: -1.0000	1st Qu.: 3559
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 22382
Mean :-0.2207	Mean :-0.2662	Mean :-0.2911	Mean : 51223
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 67091
Max. : 8.0000	Max. : 8.0000	Max. : 8.0000	Max. : 964511
BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5
Min. :-69777	Min. :-157264	Min. :-170000	Min. :-81334
1st Qu.: 2985	1st Qu.: 2666	1st Qu.: 2327	1st Qu.: 1763
Median : 21200	Median : 20088	Median : 19052	Median : 18104
Mean : 49179	Mean : 47013	Mean : 43263	Mean : 40311
3rd Qu.: 64006	3rd Qu.: 60165	3rd Qu.: 54506	3rd Qu.: 50190
Max. :983931	Max. :1664089	Max. : 891586	Max. :927171
BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3
Min. :-339603	Min. : 0	Min. : 0	Min. : 0
1st Qu.: 1256	1st Qu.: 1000	1st Qu.: 833	1st Qu.: 390
Median : 17071	Median : 2100	Median : 2009	Median : 1800
Mean : 38872	Mean : 5664	Mean : 5921	Mean : 5226
3rd Qu.: 49198	3rd Qu.: 5006	3rd Qu.: 5000	3rd Qu.: 4505
Max. : 961664	Max. :873552	Max. :1684259	Max. :896040
PAY_AMT4	PAY_AMT5	PAY_AMT6	default
Min. : 0	Min. : 0.0	Min. : 0.0	0:23364
1st Qu.: 296	1st Qu.: 252.5	1st Qu.: 117.8	1: 6636
Median : 1500	Median : 1500.0	Median : 1500.0	
Mean : 4826	Mean : 4799.4	Mean : 5215.5	
3rd Qu.: 4013	3rd Qu.: 4031.5	3rd Qu.: 4000.0	
Max. :621000	Max. :426529.0	Max. :528666.0	

```
unique(data %>% select(MARRIAGE))
```

MARRIAGE	
	1
	2
	3
	0

```
unique(data %>% select(EDUCATION))
```

EDUCATION	
	2
	1
	3
	5
	4
	6
	0

```
sum(data$MARRIAGE == 0)
```

[1] 54

```
sum(data$EDUCATION == 0)
```

```
[1] 14
```

There are no explicit missing values (NA) in the dataset. However, from the summary and checks above, we observe that some entries in MARRIAGE and EDUCATION are coded as 0, which likely represent missing or undefined categories.

Possible approaches

- Treat the value 0 in MARRIAGE and EDUCATION as a separate category (e.g., “others”).
- Impute missing values using the most frequent category (mode).

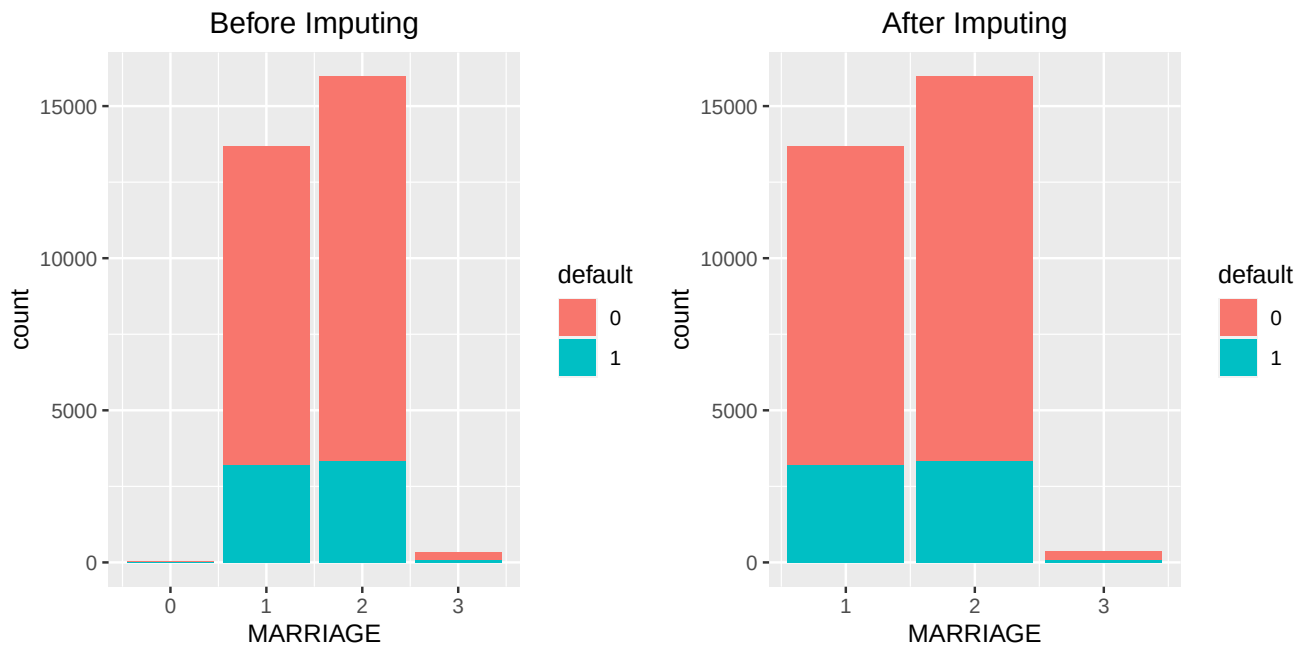
Impute the missing values

```
# Before imputation (MARRIAGE)
mplot1 <- ggplot(data = data, aes(x = MARRIAGE, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("Before Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

# Replace 0 with 3 ("others") in MARRIAGE
data$MARRIAGE <- ifelse(data$MARRIAGE == 0, 3, data$MARRIAGE)

# After imputation (MARRIAGE)
mplot2 <- ggplot(data = data, aes(x = MARRIAGE, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("After Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

grid.arrange(mplot1, mplot2, ncol = 2)
```

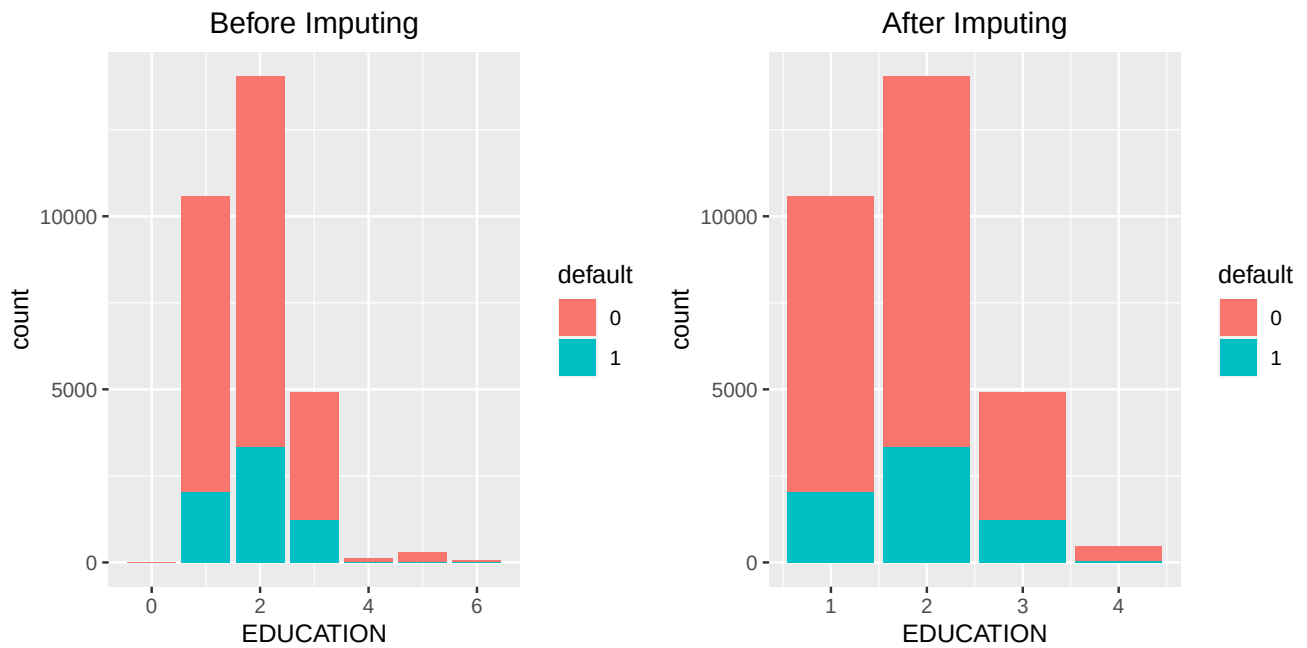


```
# Before imputation (EDUCATION)
eplot1 <- ggplot(data = data, aes(x = EDUCATION, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("Before Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

# Replace 0, 5, and 6 with 4 ("others") in EDUCATION
data$EDUCATION <- ifelse(data$EDUCATION %in% c(0, 5, 6), 4, data$EDUCATION)

# After imputation (EDUCATION)
eplot2 <- ggplot(data = data, aes(x = EDUCATION, fill = default)) +
  geom_bar() +
  stat_count(aes(label = ..count..)) +
  ggtitle("After Imputing") +
  theme(plot.title = element_text(hjust = 0.5))

grid.arrange(eplot1, eplot2, ncol = 2)
```

Solution B.2: Exploring predictors

Task: Using visualisations, explore the predictor variables to understand their distributions as well as the relationships between predictors.

Exploration of social status predictors

```
# Step 1: Calculate default rate for MARRIAGE
marriage_df <- data %>%
  group_by(MARRIAGE) %>%
  summarise(DefaultRate = mean(default == 1))

# Step 2: Calculate default rate for EDUCATION
education_df <- data %>%
  group_by(EDUCATION) %>%
  summarise(DefaultRate = mean(default == 1))

# Step 3: Calculate default rate for SEX
sex_df <- data %>%
  group_by(SEX) %>%
  summarise(DefaultRate = mean(default == 1))

# Step 4: Plot each separately

p1 <- ggplot(marriage_df, aes(x = as.factor(MARRIAGE), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(x = "Marriage", y = "Default Rate") +
  theme_minimal()

p2 <- ggplot(education_df, aes(x = as.factor(EDUCATION), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(x = "Education", y = "Default Rate") +
```

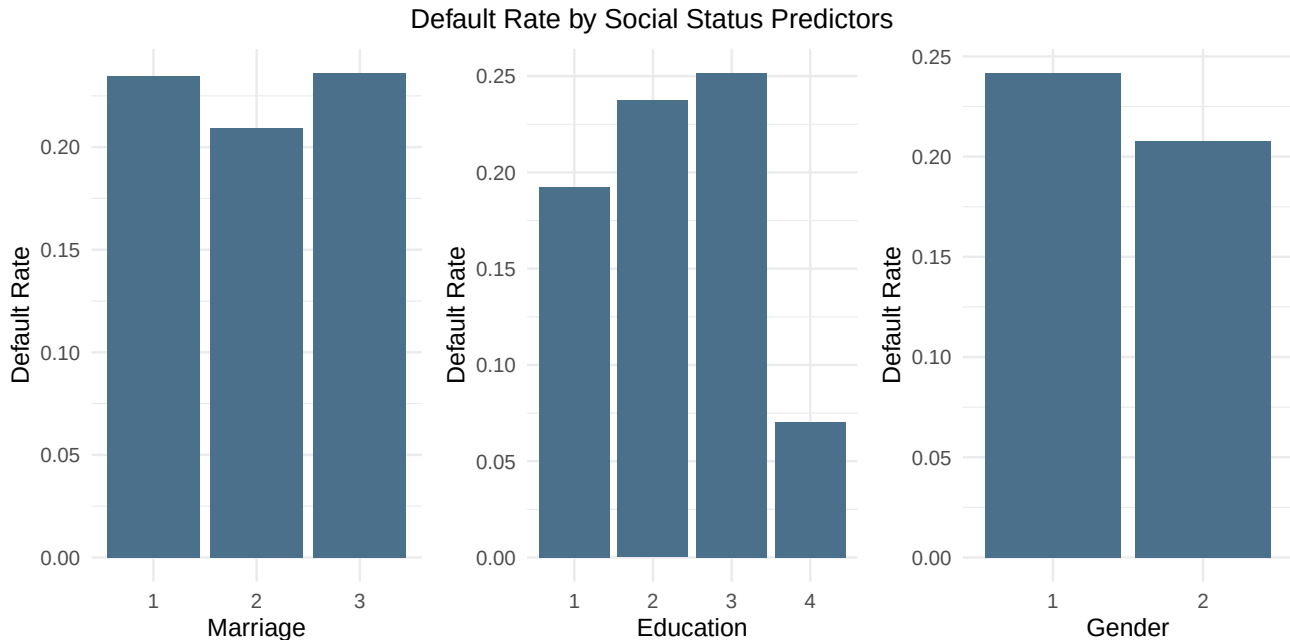
```

theme_minimal()

p3 <- ggplot(sex_df, aes(x = as.factor(SEX), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(x = "Gender", y = "Default Rate") +
  theme_minimal()

# Step 5: Arrange plots together
grid.arrange(p1, p2, p3, ncol = 3,
             top = "Default Rate by Social Status Predictors")

```



- Male customers (male = 1) appear to have a higher probability of default.
- Higher levels of education are associated with a lower probability of default.
- Married customers appear to have a higher probability of default.

Exploration of response variable

```

# Calculate the proportion of defaulters and non-defaulters
default_df <- data %>%
  group_by(default) %>%
  summarise(
    Count = n(),
    Proportion = Count / nrow(data)
  )

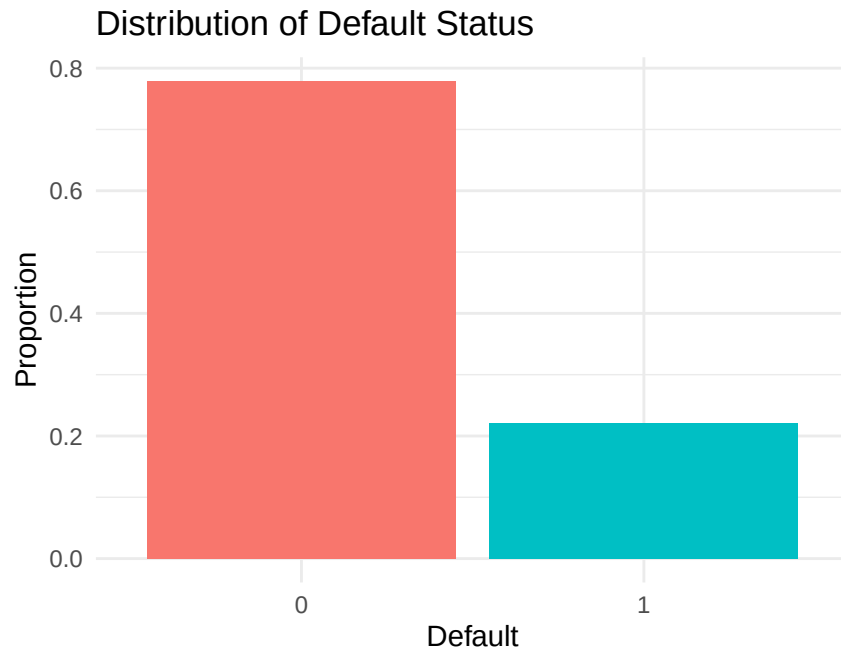
# Plot the proportion of defaulters and non-defaulters
ggplot(default_df, aes(x = default, y = Proportion, fill = default)) +
  geom_col(show.legend = FALSE) +
  labs(
    title = "Distribution of Default Status",
    x = "Default",

```

```

  y = "Proportion"
) +
theme_minimal()

```



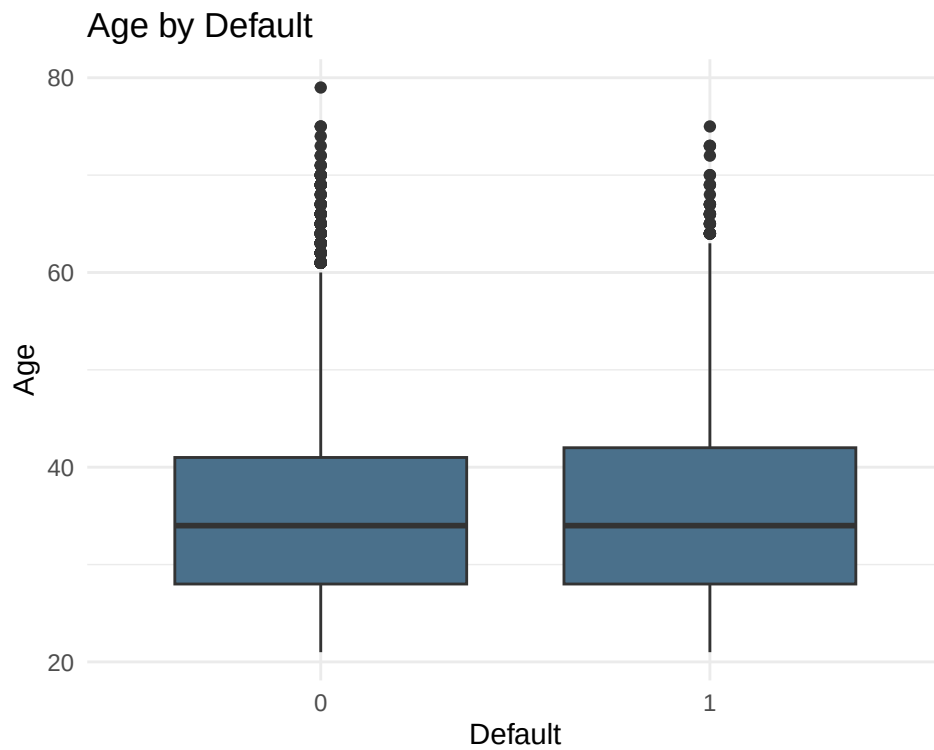
- The target variable is imbalanced, with approximately 20% defaults and 80% non-defaults. This can be addressed using under-sampling, over-sampling, or no sampling, depending on the modelling objective.

Exploration of age variable

```

# Boxplot of age by default status
ggplot(data = data, aes(x = as.factor(default), y = AGE)) +
  geom_boxplot(fill = "skyblue4") +
  labs(title = "Age by Default", x = "Default", y = "Age") +
  theme_minimal()

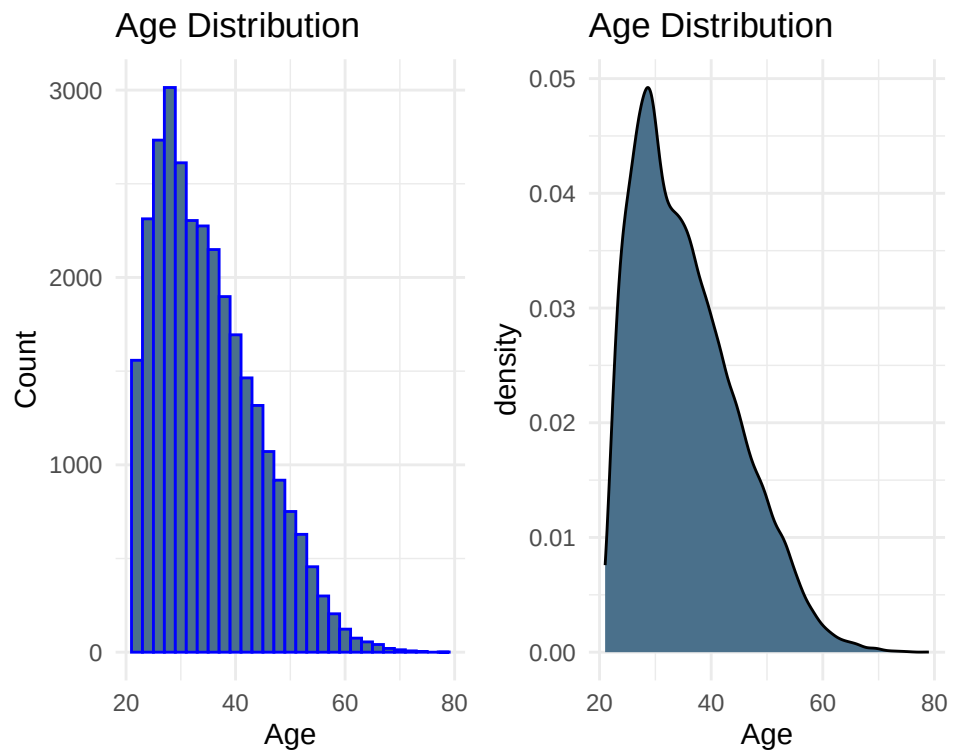
```



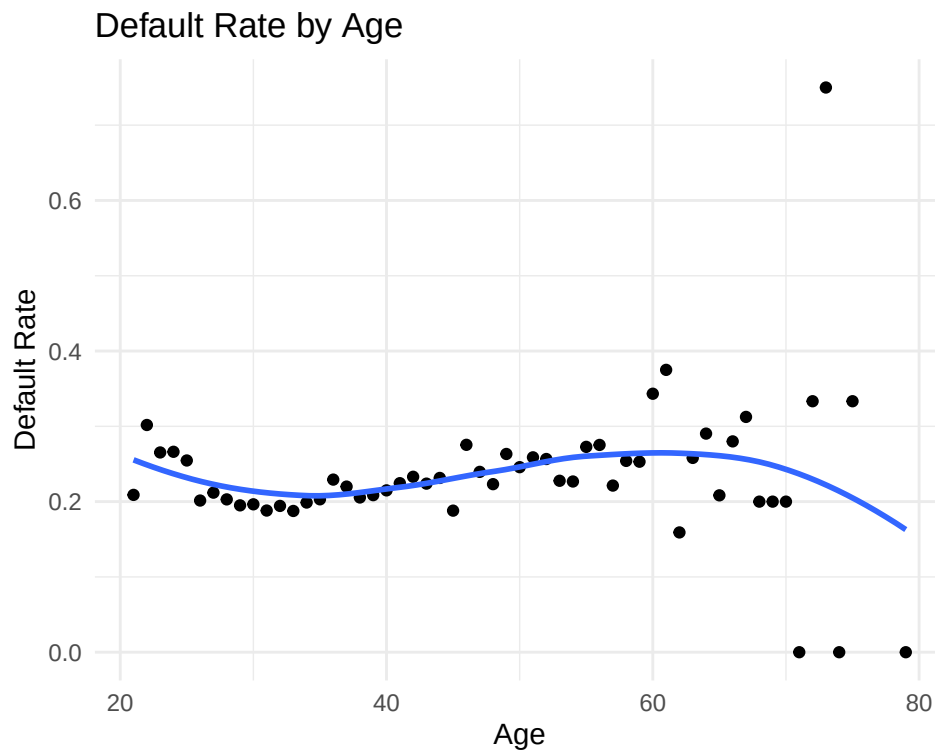
```
# Distribution of age (histogram)
plot1 <- ggplot(data, aes(x = AGE)) +
  geom_histogram(color = "blue", fill = "skyblue4") +
  labs(x = "Age", y = "Count", title = "Age Distribution") +
  theme_minimal()

# Distribution of age (density)
plot2 <- ggplot(data, aes(x = AGE)) +
  geom_density(fill = "skyblue4") +
  labs(x = "Age", title = "Age Distribution") +
  theme_minimal()

grid.arrange(plot1, plot2, ncol = 2)
```



```
# Default rate by age
data %>%
  group_by(AGE) %>%
  summarise(default_rate = mean(default == 1)) %>%
  ggplot(aes(x = AGE, y = default_rate)) +
  geom_point() +
  geom_smooth(se = FALSE) +
  labs(x = "Age", y = "Default Rate", title = "Default Rate by Age") +
  theme_minimal()
```



- In general, no clear or strong pattern is observed in the relationship between age and default rate.

Exploration of credit limit variable

```
summary(data %>% select(LIMIT_BAL))
```

```

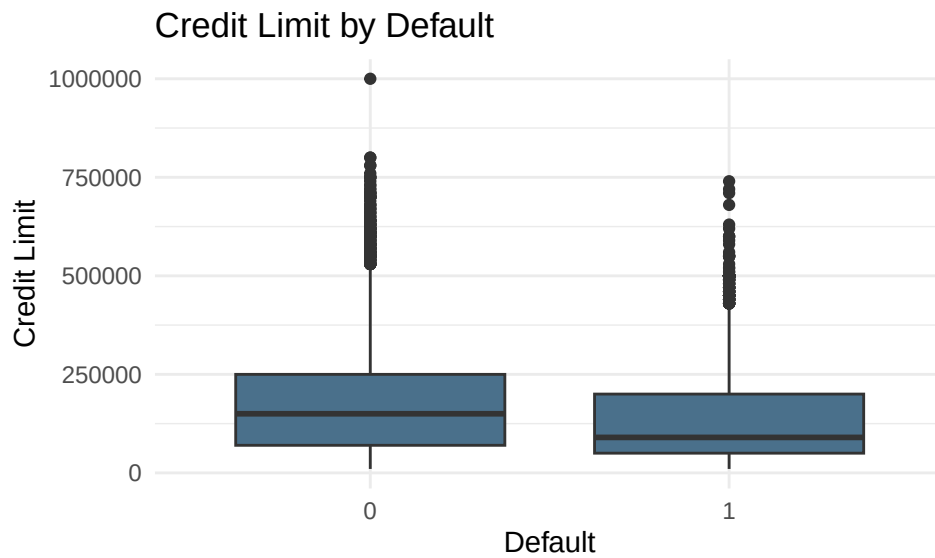
LIMIT_BAL
Min.   : 10000
1st Qu.: 50000
Median : 140000
Mean   : 167484
3rd Qu.: 240000
Max.   : 1000000

```

```

# Boxplot of credit limit by default status
ggplot(data = data, aes(x = as.factor(default), y = LIMIT_BAL)) +
  geom_boxplot(fill = "skyblue4") +
  labs(title = "Credit Limit by Default", x = "Default", y = "Credit Limit") +
  theme_minimal()

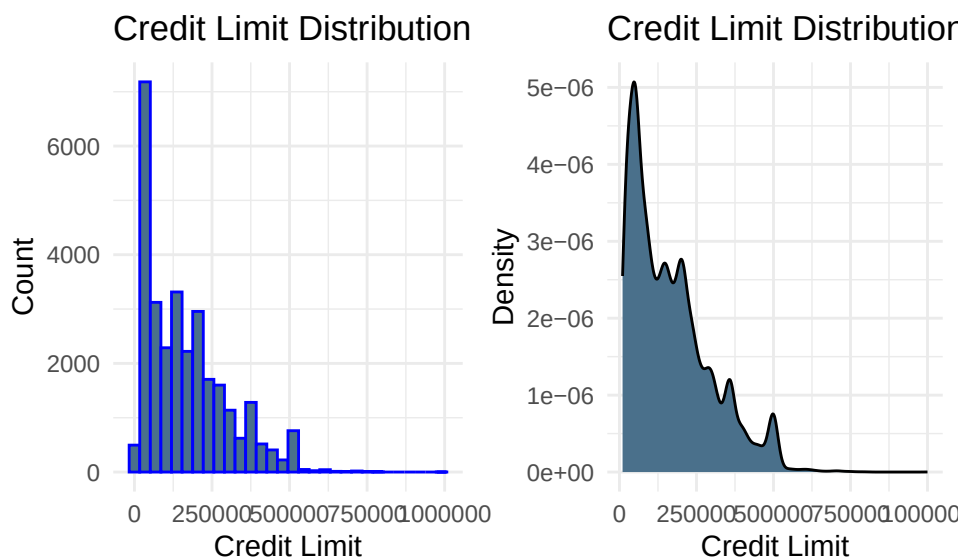
```



```
# Distribution of credit limit (histogram)
plot_bal1 <- ggplot(data, aes(x = LIMIT_BAL)) +
  geom_histogram(color = "blue", fill = "skyblue4") +
  labs(title = "Credit Limit Distribution", x = "Credit Limit", y = "Count") +
  theme_minimal()

# Distribution of credit limit (density)
plot_bal2 <- ggplot(data, aes(x = LIMIT_BAL)) +
  geom_density(fill = "skyblue4") +
  labs(title = "Credit Limit Distribution", x = "Credit Limit", y = "Density") +
  theme_minimal()

grid.arrange(plot_bal1, plot_bal2, ncol = 2)
```



- Lower credit limits appear to be associated with a higher probability of default.

Exploration of bill statement amount variables

```
billamt_colnames <- paste0("BILL_AMT", 1:6)
```

```
bill_data <- data %>%  
  select(starts_with("BILL_AMT"))
```

```
summary(bill_data)
```

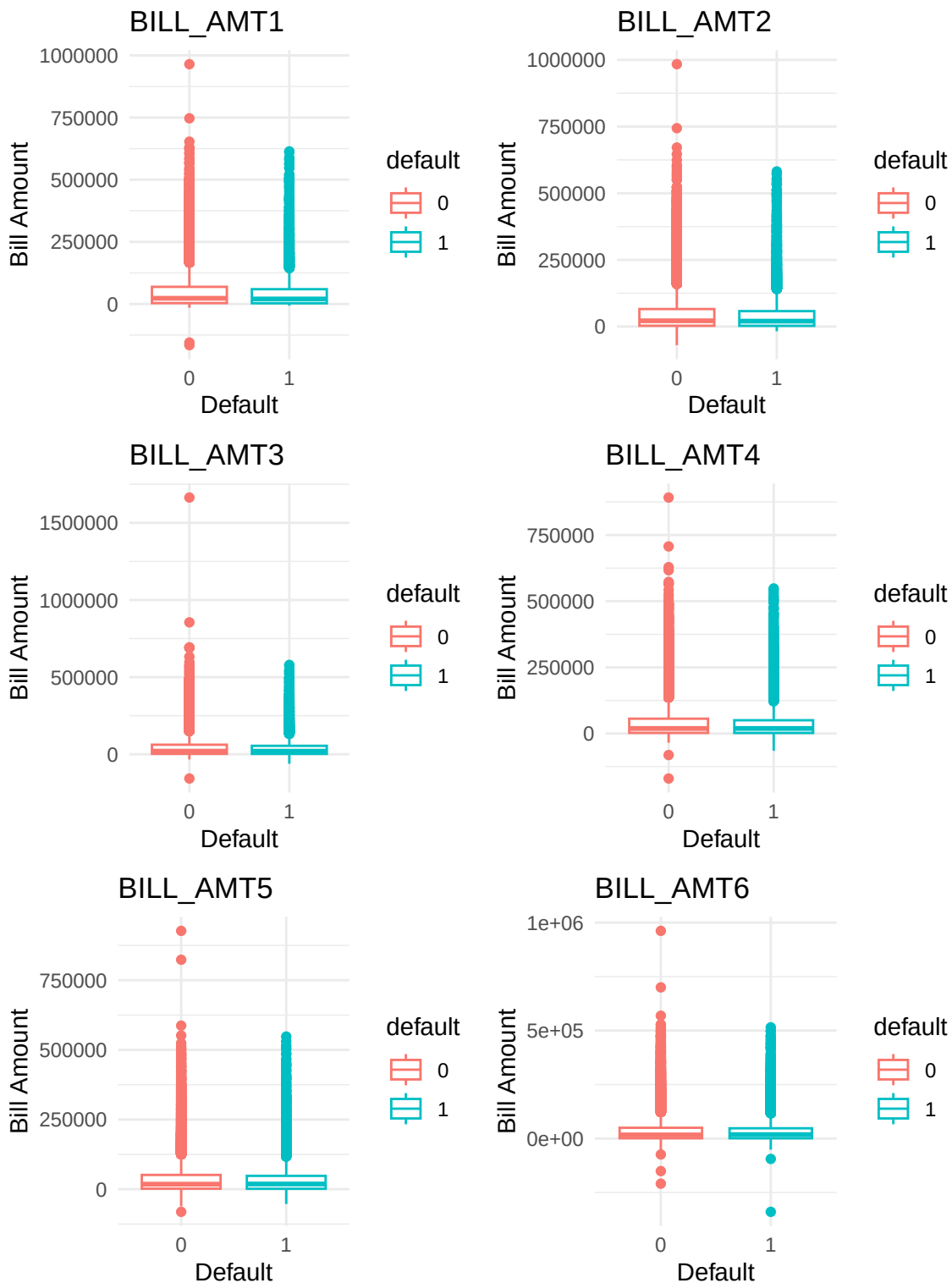
BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4
Min. : -165580	Min. : -69777	Min. : -157264	Min. : -170000
1st Qu.: 3559	1st Qu.: 2985	1st Qu.: 2666	1st Qu.: 2327
Median : 22382	Median : 21200	Median : 20088	Median : 19052
Mean : 51223	Mean : 49179	Mean : 47013	Mean : 43263
3rd Qu.: 67091	3rd Qu.: 64006	3rd Qu.: 60165	3rd Qu.: 54506
Max. : 964511	Max. : 983931	Max. : 1664089	Max. : 891586

BILL_AMT5	BILL_AMT6
Min. : -81334	Min. : -339603
1st Qu.: 1763	1st Qu.: 1256
Median : 18104	Median : 17071
Mean : 40311	Mean : 38872
3rd Qu.: 50190	3rd Qu.: 49198
Max. : 927171	Max. : 961664

```
# Boxplots of bill statement amounts by default status
```

```
boxplots <- lapply(1:ncol(bill_data), function(i) {  
  ggplot(data = data, aes(x = default, y = bill_data[[i]], colour = default)) +  
    geom_boxplot() +  
    labs(  
      x = "Default",  
      y = "Bill Amount",  
      title = billamt_colnames[i]  
    ) +  
    theme_minimal()  
})
```

```
do.call(grid.arrange, c(boxplots, ncol = 2, nrow = 3))
```

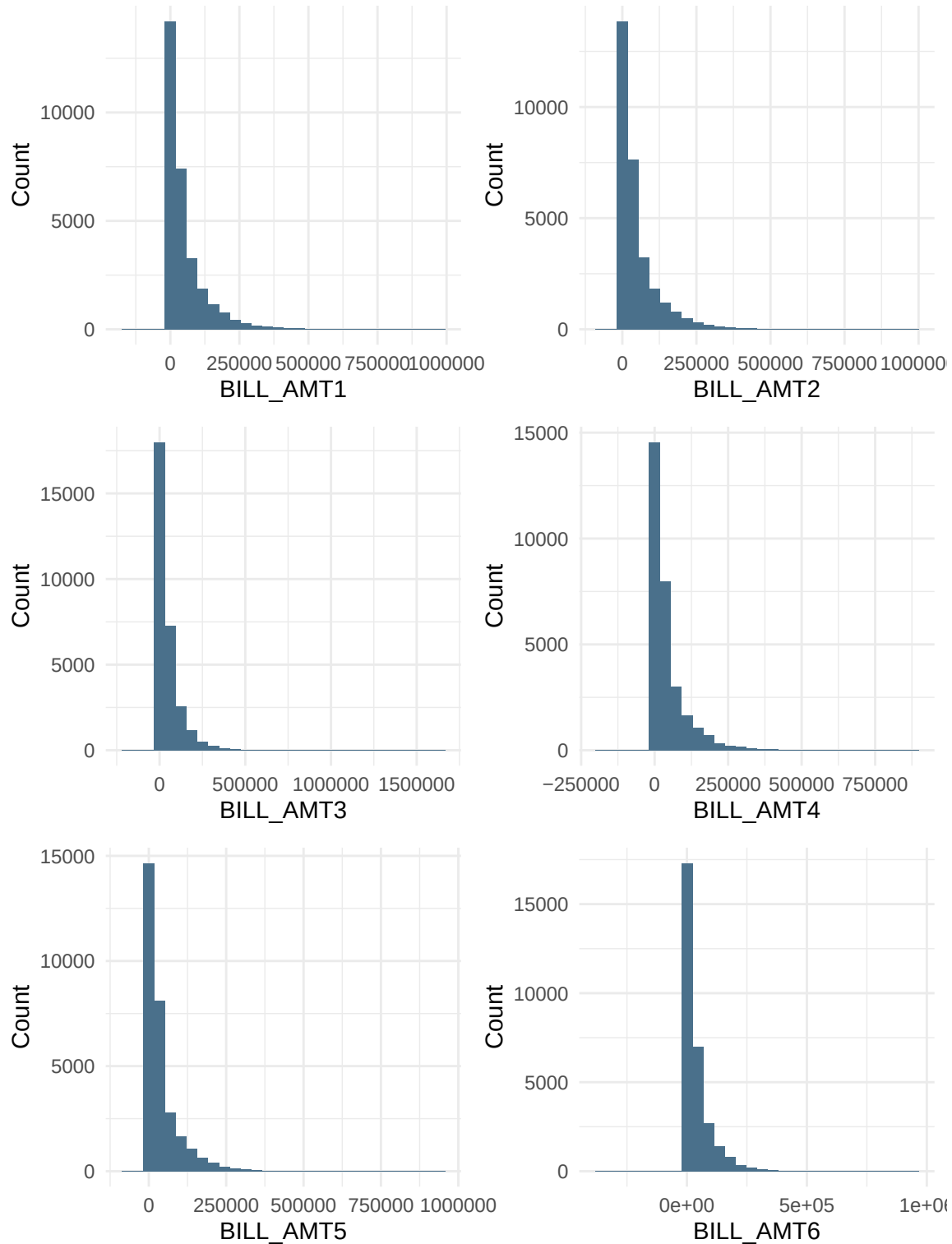
```
# Histograms of bill statement amounts
histograms <- lapply(1:ncol(bill_data), function(i) {
  ggplot(data = bill_data, aes(x = bill_data[[i]])) +
    geom_histogram(fill = "skyblue4") +
    labs(
      x = billamt_colnames[i],
      y = "Count"
    )
})
```

```

) +
  theme_minimal()
})

do.call(grid.arrange, c(histograms, ncol = 2, nrow = 3))

```



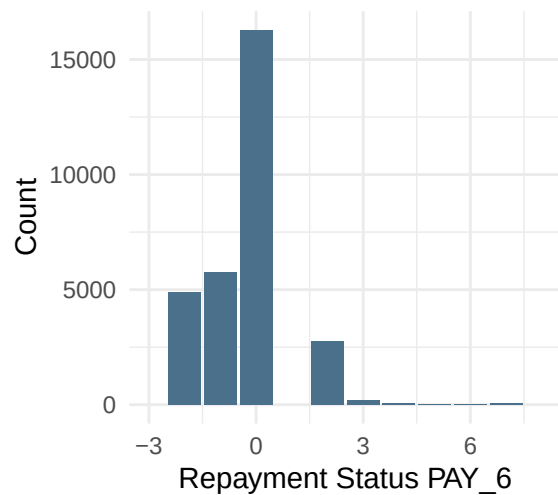
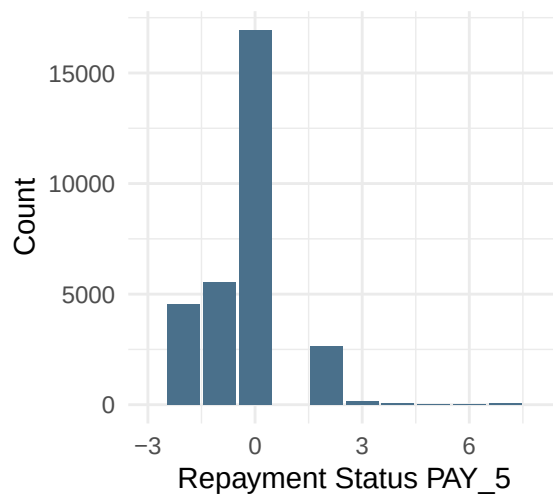
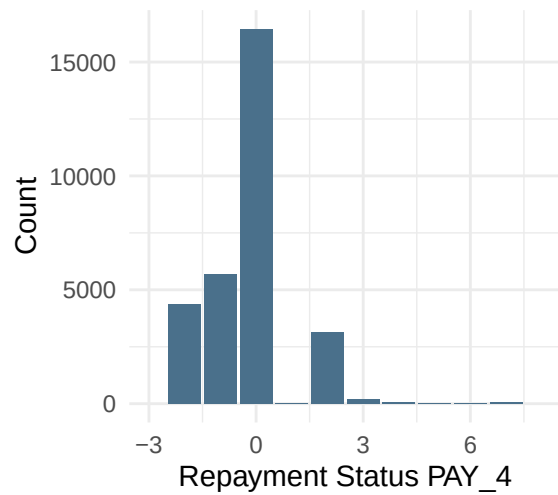
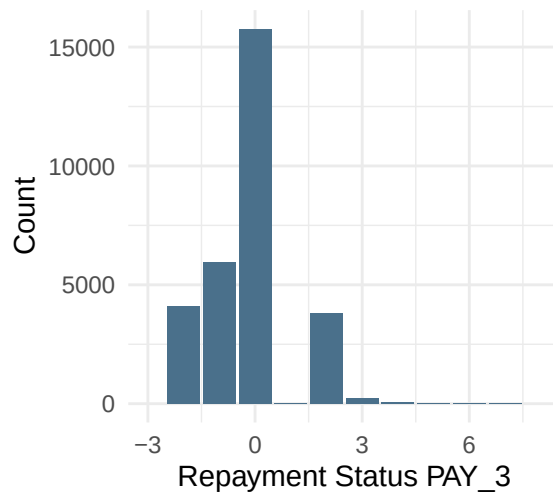
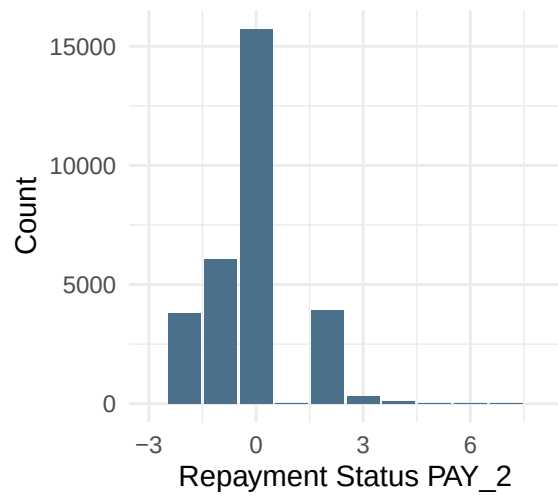
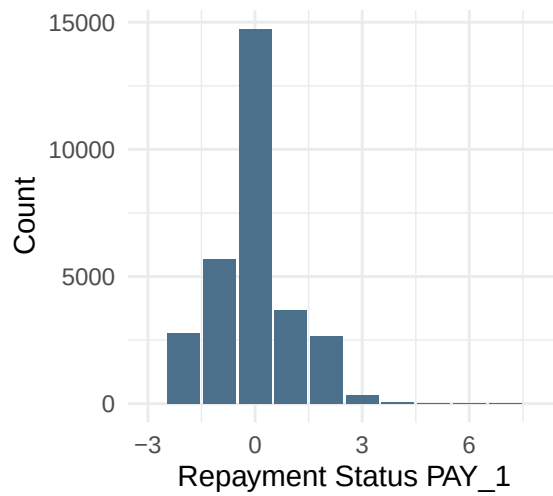
- In general, we observe a decreasing trend in the key summary statistics from BILL_AMT1 to BILL_AMT6.

Exploration of history of past payment variables

```
pay_colnames <- paste0("PAY_", 1:6)
pay_data <- data %>% select(all_of(pay_colnames))

# Distribution of repayment status (bar plots)
barplots <- lapply(1:ncol(pay_data), function(i) {
  ggplot(pay_data, aes(x = pay_data[[i]])) +
    geom_bar(fill = "skyblue4") +
    labs(
      x = paste("Repayment Status", pay_colnames[i]),
      y = "Count"
    ) +
    xlim(-3, 8) +
    theme_minimal()
})

do.call(grid.arrange, c(barplots, ncol = 2, nrow = 3))
```



```
# Default rate by repayment status
default_plots <- lapply(1:ncol(pay_data), function(i) {

  temp_df <- data.frame(
    Status = pay_data[[i]],
    default = data$default
  )
})
```

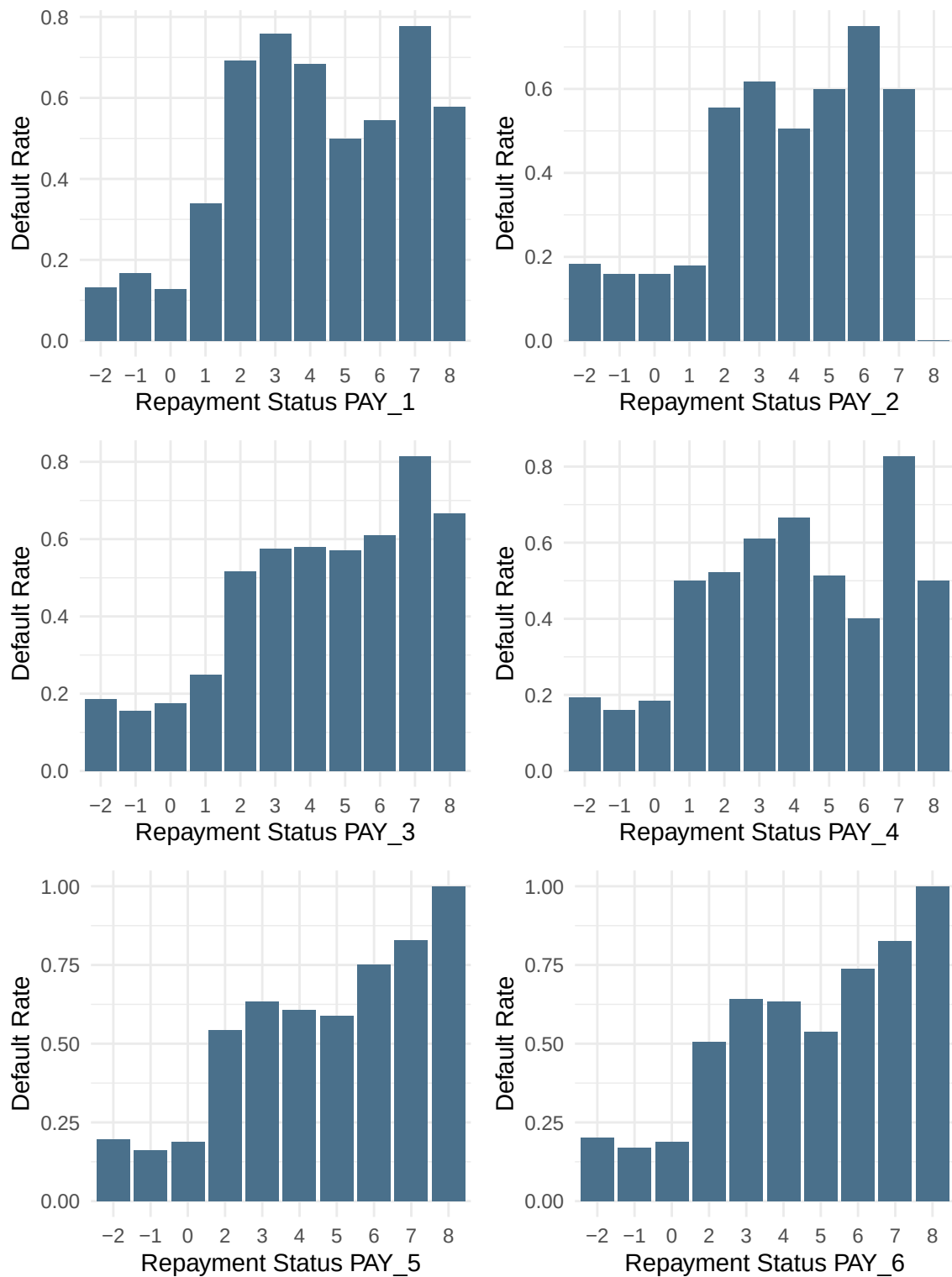
```

temp_summary <- temp_df %>%
  group_by(Status) %>%
  summarise(
    DefaultRate = mean(default == 1),
    .groups = "drop"
  )

ggplot(temp_summary, aes(x = as.factor(Status), y = DefaultRate)) +
  geom_col(fill = "skyblue4") +
  labs(
    x = paste("Repayment Status", pay_colnames[i]),
    y = "Default Rate"
  ) +
  theme_minimal()
})

do.call(grid.arrange, c(default_plots, ncol = 2, nrow = 3))

```



- Payment delays, even for one month, are associated with a higher probability of default.

Solution B.3: Transformations

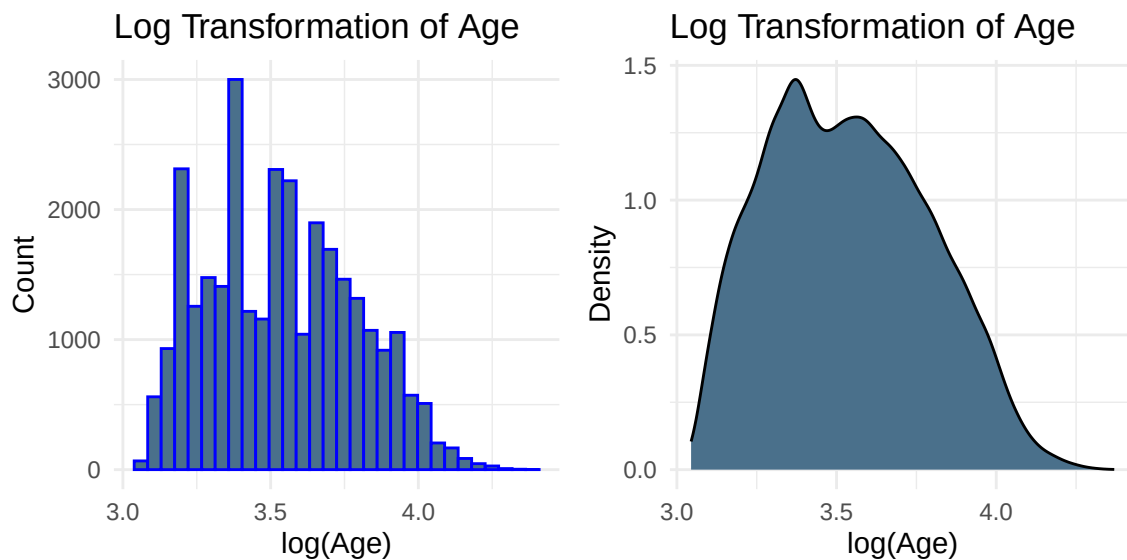
Task: Are there any transformations of the predictors that might improve model performance?

Relevant transformations of predictors

```
# Log transformation of AGE
```

```
plot3 <- ggplot(data, aes(x = log(AGE))) +  
  geom_histogram(colour = "blue", fill = "skyblue4") +  
  labs(title = "Log Transformation of Age", x = "log(Age)", y = "Count") +  
  theme_minimal()  
  
plot4 <- ggplot(data, aes(x = log(AGE))) +  
  geom_density(fill = "skyblue4") +  
  labs(title = "Log Transformation of Age", x = "log(Age)", y = "Density") +  
  theme_minimal()
```

```
grid.arrange(plot3, plot4, ncol = 2)
```

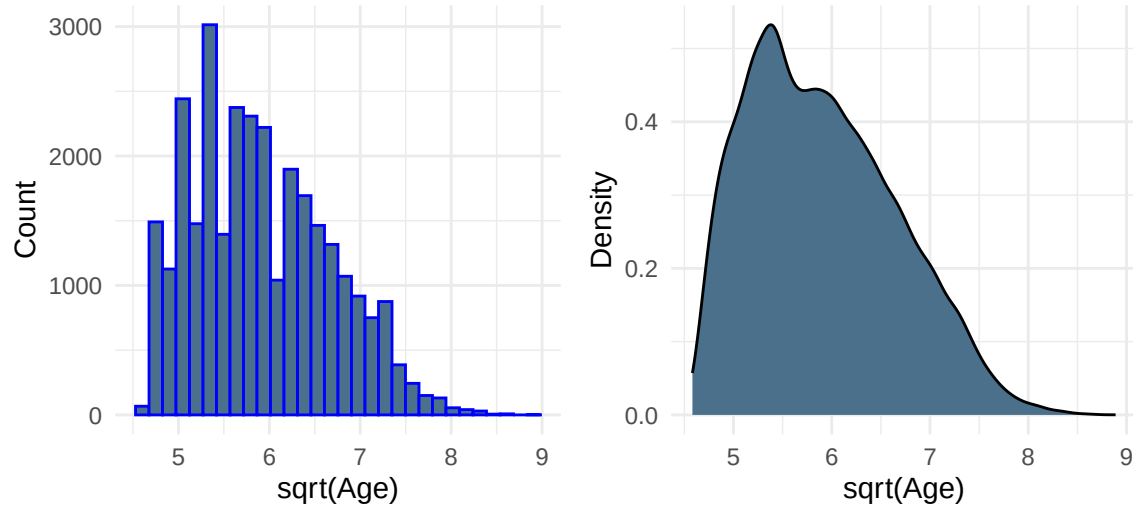


```
# Square-root transformation of AGE
```

```
plot5 <- ggplot(data, aes(x = sqrt(AGE))) +  
  geom_histogram(colour = "blue", fill = "skyblue4") +  
  labs(title = "Square-root Transformation of Age", x = "sqrt(Age)", y = "Count") +  
  theme_minimal()  
  
plot6 <- ggplot(data, aes(x = sqrt(AGE))) +  
  geom_density(fill = "skyblue4") +  
  labs(title = "Square-root Transformation of Age", x = "sqrt(Age)", y = "Density") +  
  theme_minimal()
```

```
grid.arrange(plot5, plot6, ncol = 2)
```

Square-root Transformation of Age Square-root Transformation of



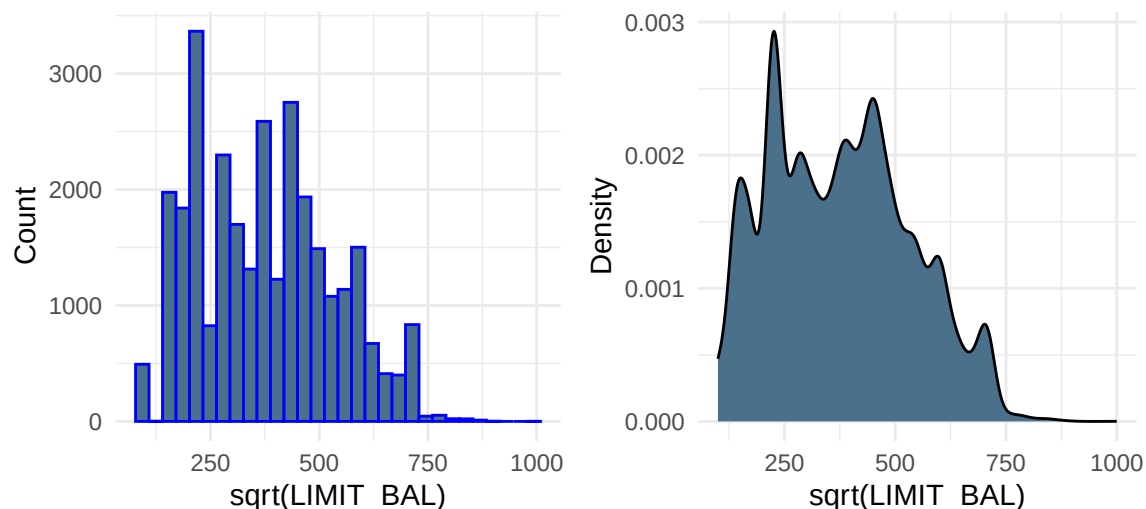
```
# Square-root transformation of LIMIT_BAL
```

```
plot_bal3 <- ggplot(data, aes(x = sqrt(LIMIT_BAL))) +  
  geom_histogram(colour = "blue", fill = "skyblue4") +  
  labs(title = "Square-root Transformation of Credit Limit", x = "sqrt(LIMIT_BAL)", y = "Count") +  
  theme_minimal()
```

```
plot_bal4 <- ggplot(data, aes(x = sqrt(LIMIT_BAL))) +  
  geom_density(fill = "skyblue4") +  
  labs(title = "Square-root Transformation of Credit Limit", x = "sqrt(LIMIT_BAL)", y = "Density") +  
  theme_minimal()
```

```
grid.arrange(plot_bal3, plot_bal4, ncol = 2)
```

Square-root Transformation of Credit Square-root Transformation c



- While transformations such as logarithmic and square-root transformations are commonly used to reduce skewness, the visual evidence here is not strong enough to suggest that they are necessary.
- In practice, transformations can be treated as part of the modelling process. For example, we may compare models built using the original variables and their transformed versions, and evaluate their performance using validation data.

Solution B.4: Correlation and multicollinearity

Task: Rename the column “default payment next month” as “default”. Are there strong relationships between the default variable and other numeric variables? How can you handle highly correlated variables?

Relationships between the default variable and other numeric variables

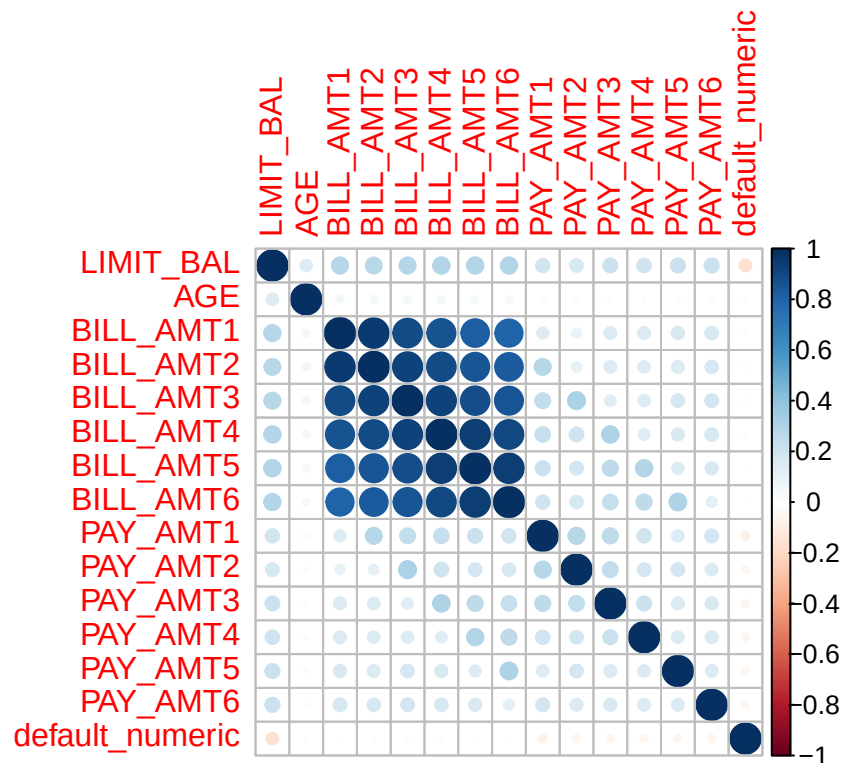
Here, we examine the correlation between the default variable and other numeric variables.

```
# Select variables to exclude from the correlation matrix
pay_colnames <- paste0("PAY_", 1:6)

# Create a numeric version of the default variable for correlation analysis
data$default_numeric <- as.numeric(as.character(data$default))

# Correlation plot
cor_data <- data %>%
  select(-EDUCATION, -SEX, -MARRIAGE, -ID, -all_of(pay_colnames), -default) %>%
  select(where(is.numeric))

corrplot(cor(cor_data), method = "circle")
```



- We observe strong linear correlations between the bill statement amounts in different months. This is expected because these variables describe related financial quantities measured repeatedly over time.
- In the presence of multicollinearity, possible approaches include ridge regression, lasso regression, elastic net regression, and principal component analysis (PCA). Removing variables is also a possible option, but it is typically considered a last resort, as it may discard useful information and reduce interpretability.

Principal component analysis

```
# PCA

pca_model <- prcomp(
  cor_data,
  center = TRUE,
  scale. = TRUE
)

summary(pca_model)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.4333	1.3235	1.02473	1.00038	0.95589	0.93941	0.93376
Proportion of Variance	0.3947	0.1168	0.07001	0.06672	0.06092	0.05883	0.05813
Cumulative Proportion	0.3947	0.5115	0.58151	0.64823	0.70915	0.76798	0.82611

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.88285	0.8521	0.82363	0.51373	0.26648	0.20260	0.15919
Proportion of Variance	0.05196	0.0484	0.04522	0.01759	0.00473	0.00274	0.00169
Cumulative Proportion	0.87807	0.9265	0.97170	0.98929	0.99402	0.99676	0.99845

	PC15
Standard deviation	0.15244
Proportion of Variance	0.00155
Cumulative Proportion	1.00000

References

- Noll, Alexander, Robert Salzmänn, and Mario V Wuthrich. 2020. "Case Study: French Motor Third-Party Liability Claims." *Available at SSRN 3164764*.
- Yeh, I-Cheng, and Che-hui Lien. 2009. "The Comparisons of Data Mining Techniques for the Predictive Accuracy of Probability of Default of Credit Card Clients." *Expert Systems with Applications* 36 (2): 2473–80.